

SHOWCASE

Raspberry Pi “High Quality Camera”

The Raspberry Pi’s camera module has come a long way from its 2016 model, the “Camera Module V2”. Dubbed the “High Quality Camera”, this module is better than its previous iteration in every way, except for the fact that it’s not as compact, understandably. The new camera module is ideal for budding photographers who want to work with cameras and integrate it into their projects. It comes with



a 7.9mm Sony IMX477 12.3-MP back-illuminated sensor and a mount that works with C- and CS-mount lenses.

The High Quality Camera doesn’t come with a lens, however, Raspberry Pi resellers will have a 6mm CS-mount CCTV lens’ for sale at \$15 and a high-quality 16mm C-mount lens’ for \$50. The Raspberry Pi High Quality Camera module is available for \$50, along with a dust cap, C-CS adapter, tripod mount, and a ribbon cable to connect to the Raspberry Pi. This setup can be taken even further, allowing those interested to adapt their 16mm mount to a modern APS-C or even a full-frame lens!



Boeing Loyal Wingman

The newest addition to the Royal Australian Air Force (RAAF) is Boeing’s combat AI drone, the Loyal Wingman. Drones have been a part of the military for a while now, but the Loyal Wingman isn’t like your conventional drone. First off, it’s pretty large, large enough to accompany a manned aircraft. The drone is piloted by its AI, and isn’t remotely controlled like other drones. This doesn’t mean that there’s no manual control, there is, and it’s apparently incredibly user-friendly and doesn’t use a bulky remote controller which is usually the norm when it comes to military drones. The Loyal Wingman is still in its prototype stages, and the RAAF is currently carrying out flight tests and demos to see how the Loyal Wingman integrates with other drones in the force, and also to test how safe it is to have an unmanned AI-controlled drone in the skies; the risk to pilots in the same airspace. The Loyal Wingman’s first flight is expected to happen later this year, and if it’s a success, it could mean we’d see a lot more unmanned aircraft drones in Air Forces around the world.

R9-D9 Vacuum

Being stuck at home sure can bring the creative side out in some people. One Matthew Scott Hunter for example, managed to create his own, functional, robot vacuum Artoo unit that roams about cleaning his house. The R9-D9’s inspiration is very obvious, heck, the reason it’s the R9-D9 and not the R2-D2 is because he couldn’t replicate the astromech exactly.

The droid’s body is made using a garbage can, and a domed head. The legs were made using a stiff poster



board, and there are a bunch of other household items that have been thrown in there to give the droid its look. Like we mentioned above, it didn’t look identical to the original,

so Matthew went with a different look as well in terms of the paint-job, giving the R9-D9 a more Iron-Man inspired colour-scheme.

Overall, the robot vacuum weighs roughly 7kgs. The light weight meant it didn’t impede the robot vacuum it was attached to too much and didn’t eat through the battery too fast. However, the larger size most definitely means it’s not going to be cleaning a lot of those hard to reach places. Kind of defeats the purpose but who cares?